PRPrompt: Prototype-Routed Graph Prompting for Few-Shot Node Classification

Abstract

We introduce PRPrompt, a graph prompting method that uses support-set prototypes to route trainable prompts to individual nodes. With only a few labeled nodes per class, adaptation must determine both what a prompt should encode and where its contribution should be applied. PRPrompt constructs class references from aggregated node attributes, refines them using unlabeled graph features while retaining support anchors, and converts node–prototype associations into a fixed routing matrix. The resulting combinations of class prompt tokens are injected into a frozen graph convolutional encoder, followed by prototype classification. We evaluate each 1-, 3-, and 5-shot setting separately on Cora, CiteSeer, and PubMed using matched support draws and training seeds. In the 5-shot setting, PRPrompt achieves accuracies of 73.02%, 66.17%, and 70.98%, respectively; replacing a shared prompt with prototype routing improves accuracy by 36.94, 40.45, and 3.02 percentage points in the corresponding conditions. These paired results support the value of prototype-guided prompt allocation within the evaluated transductive setting.

1 Introduction

Pretrained graph neural networks offer a starting point for learning with scarce node labels. Graph prompting adapts these representations through a small set of task parameters, using feature modifications, prompt structures, or task-specific readout [3, 10, 14, 15]. Beyond choosing an injection point, an adaptation method must decide how a prompt's contribution is distributed across nodes. This allocation matters because nodes in the same graph can play different roles in the downstream classification task.

A shared prompt expresses a global adjustment. More flexible methods learn input-dependent mixtures, assign prompts to subgraphs, or select nodes through a policy [3, 8, 22]. These approaches establish that node-dependent allocation is not itself a new idea. Our question is more specific: *can the few labeled examples already available for a task define useful routing coefficients, so that optimization can focus on prompt content?* Support examples provide class references that can be constructed before adaptation and used without training an additional routing network.

We propose PRPrompt, which uses support-anchored prototypes as coordinates for prompt allocation. Nodes with similar class associations receive related combinations of class prompt tokens. The routing representation comes from aggregated graph attributes; its prototypes are initialized by labeled support nodes and refined with unlabeled features while retaining the support anchors. These associations yield a fixed routing matrix that mixes trainable tokens inside a frozen GCN. Only the prompt tokens are optimized, and prediction uses a prototype readout. Figure 1 summarizes this separation between fixed allocation and learned content.

Our evaluation asks two questions. First, how does complete adaptation compare with frozen pretraining and full fine-tuning in matched local runs, and where does its accuracy lie relative to literature-reported prompt results? Second, does prototype routing add value over a shared prompt when the injection point and training recipe are held fixed? Local runs cover Cora, CiteSeer, and PubMed at 1, 3, and 5 shots. External accuracy values come from Chen et al. [2] and retain their original protocol. Every result remains specific to its dataset and shot count. The gain over frozen pretraining is much smaller on PubMed than on Cora or CiteSeer.

The paper makes three contributions:

- A graph prompting construction that uses support-anchored class relationships to fix node-specific allocation before prompt optimization.
- A hidden-layer implementation with $C \times 128$ trainable prompt parameters and no learned routing network.
- Matched adaptation comparisons and shared-prompt ablations, with seed variability and paired uncertainty reported separately for each evaluated condition.

The contribution is the use of prototype relationships to organize hidden-layer prompts; it does not treat prototypes or node-dependent prompting as new primitives.

2 Related Work

Graph representations and pretraining. Graph convolution, neighborhood sampling, and graph attention provide complementary ways to construct node representations [4, 7, 16]. Self-supervised objectives include local–global information maximization [17], augmentation-based contrastive learning [18], and masked reconstruction [5]. Pretraining objectives can affect downstream transfer, and combining node- and graph-level objectives has been studied explicitly [6]. Our study fixes the pretrained checkpoint

on each graph and examines downstream adaptation, rather than proposing a new pretraining objective.

Aligning pretraining with downstream tasks. GPPT and GraphPrompt connect pretraining to downstream prediction through task reformulation and prompting [10, 14]. All in One addresses multiple task levels with prompt tokens, structures, and insertion rules [15]; MultiGPrompt combines multiple pretext tasks and downstream prompts [20]. HGPrompt and HetGPT extend prompting to heterogeneous graph settings [11, 19]. These methods address task or graph-type alignment. We focus on a narrower design question within few-shot node classification: how task-class references should allocate prompts across nodes.

Adaptive prompt allocation. GPF-plus learns input-dependent prompt combinations [3], SUPT assigns prompts at subgraph level [8], and RELIEF learns node selection together with prompt content [22]. ProNoG studies conditional node-specific prompts for non-homophilic graphs [21], while DAGPrompT addresses distribution-aware adaptation [1]. Attribute-guided dynamic prompting has also been studied for robustness to graph perturbations [9]. In contrast to learning an allocation function, PRPrompt computes support-conditioned prototype relationships once and keeps the resulting route fixed. These works delimit the design space; they are not all experimental baselines in our controlled comparison.

Class prototypes and evaluation. Prototypical Networks use support-set class representatives for prediction [13], and semi-supervised extensions incorporate unlabeled examples into prototype construction [12]. We use these established ideas to obtain task references for routing. Routing prototypes live in aggregated attribute space; prediction prototypes come from the adapted encoder output. ProG [23] motivates explicit evaluation protocols. Chen et al. [2] report adaptation results across pretraining strategies; we reuse their baseline accuracy values as literature context while retaining local matched controls for the routing claim.

3 Problem Setting

Let $G = (V, E, X)$ be an attributed graph with n nodes, features $X \in \mathbb{R}^{n\times d}$, and C task classes. For each class c, a support set S_c contains k labeled nodes, with $k \in \{1, 3, 5\}$. Let $S = \bigcup_c S_c$ and let T be a disjoint query set. The full graph topology and all node features are visible; only labels in S are provided to fitting and hyperparameter selection. Query labels are used for scoring after predictions are sealed.

The encoder is a pretrained two-layer GCN [7] with hidden and output dimensions of 128. Its parameters are frozen for PRPrompt. The task is transductive, same-graph classification over known classes: query features participate in graph aggregation, prototype refinement, and global centering. It is not an inductive unseen-class evaluation. Throughout the paper, a *condition* fixes the dataset, shot count, method, and pretrained checkpoint.

4 Prototype-Routed Prompting

Overview. The method separates allocation from content. First, support-anchored prototypes define class relationships in aggregated attribute space. Second, a fixed routing matrix turns those relationships into node-specific combinations of prompts. Third, support supervision learns the prompt tokens inside the frozen encoder. The following subsections follow this computation order.

4.1 Support-anchored class references

Let $\widehat{A} = \widetilde{D}^{-1/2}(A + I)\widetilde{D}^{-1/2}$ denote the normalized adjacency with self-loops. We form a routing representation from two-hop aggregated attributes:

$$\begin{aligned} X^{(2)} &= \widehat{A}^2 X, \\ R &= \mathrm{rownorm}\Big(X^{(2)} - \mathbf{1}\,\mathrm{mean}_{v\in V}\, X_v^{(2)}\Big). \end{aligned} \tag{1}$$

Here rownorm applies Euclidean normalization to each row. Routing therefore uses graph attributes and structure, rather than a separate learned network. Initial class references are

$$\mu_c^{(0)} = \frac{1}{k}\sum_{i\in S_c} R_i. \tag{2}$$

To incorporate the unlabeled graph while retaining a labeled reference, let $U_V = V \setminus S$. Starting from $\mu^{(0)}$, we use $i \in U_V$ and $t = 0, 1, 2$ for three fixed refinement iterations:

$$a_{ic}^{(t)} = \frac{\exp(R_i^\top \mathrm{norm}\Big(\mu_c^{(t)}\Big)/0.05)}{\sum_{c'}\exp(R_i^\top \mathrm{norm}\Big(\mu_{c'}^{(t)}\Big)/0.05)}, \tag{3}$$

$$\mu_c^{(t+1)} = 0.5\mu_c^{(0)} + 0.5\frac{\sum_{i\in U_V} a_{ic}^{(t)} R_i}{\max(\sum_{i\in U_V} a_{ic}^{(t)}, 10^{-8})}. \tag{4}$$

The anchor is always the original support prototype, not the previous iterate. This is a fixed prototype construction; we do not claim an expectation-maximization objective or a convergence guarantee. No labels from U_V enter these computations.

4.2 From class relationships to a fixed routing matrix

Using the refined prototypes, we compute node–class associations at temperature $\tau_r = 0.1$:

$$\begin{aligned} Q_{ic} &= \mathrm{softmax}_c\Big(R_i^\top \mathrm{norm}\Big(\mu_c^{(3)}\Big)/\tau_r\Big), \\ B &= \widehat{A}\,(Q - \mathbf{1}\,\mathrm{mean}_{v\in V}\, Q_v). \end{aligned} \tag{5}$$

Centering expresses a node's class association relative to the graph-wide average; neighborhood aggregation

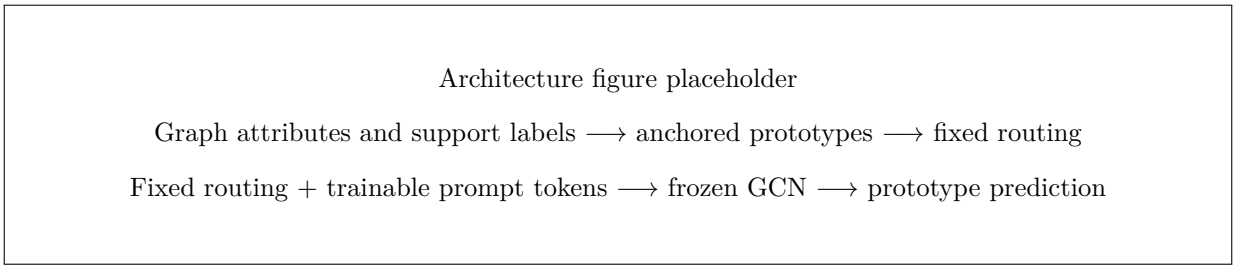


Figure 1: Overview of PRPrompt. Support labels initialize class references in aggregated attribute space. Unlabeled graph features refine these references, and the resulting node–prototype associations determine a fixed routing matrix. Only the class prompt tokens are optimized; the routed prompts enter the first hidden layer of a frozen GCN. Query labels are excluded from fitting and selection. The final architecture artwork is intentionally left as a placeholder.

incorporates adjacent association patterns. Q is a row-normalized association matrix, whereas B contains signed prompt-combination coefficients. We do not interpret B as a class-probability matrix. B is computed once per support set and kept fixed throughout training.

4.3 Adapting frozen hidden computations

Let H^{pre} be the first GCN layer's preactivation on the original graph. We learn a zero-initialized matrix $P \in \mathbb{R}^{C\times 128}$. With s the vector of channelwise sample standard deviations of H^{pre}, clamped below by 10^{-6}, the prompted hidden representation is

$$\widetilde{H}^{\text{pre}} = H^{\text{pre}} + 10\,[B\tanh(P)] \odot s. \tag{6}$$

The scale vector is broadcast over nodes. The existing LeakyReLU and second graph convolution then produce Z. Thus, prompt content is learned through support supervision, while routing and backbone parameters remain fixed. Zero initialization recovers the frozen encoder at the start of optimization. The trainable prompt counts are 896, 768, and 384 for Cora, CiteSeer, and PubMed.

4.4 Training and prediction

We optimize the prompt matrix for 100 Adam steps with learning rate 0.01 and no weight decay:

$$\mathcal{L}(P) = \frac{1}{|S|}\sum_{i\in S}\mathrm{CE}(\ell_i(P), y_i) + 10^{-3}\,\mathrm{mean}(P\odot P). \tag{7}$$

The support logits ℓ_i use cosine prototype classification. The same prototype readout is kept throughout the routing ablation. Support leave-one-out construction, prediction normalization, and temperatures are specified in Appendix A.

The routing is a one-time computation. For sparse graphs, the two attribute aggregation steps cost $O(|E|d)$; prototype refinement costs $O(nCd)$ per iteration; and multiplying B by prompt tokens costs $O(nC\cdot 128)$ per optimization step, in addition to the frozen encoder's forward and backward computations. Freezing weights reduces trainable parameters but does not eliminate gradient computation through the second convolution. We report parameter counts, not an unmeasured runtime advantage.

5 Experiments

5.1 Protocol and comparison methods

Our local evaluation covers Cora, CiteSeer, and PubMed at 1, 3, and 5 shots per class. Official training masks define support pools and official test masks define fixed query sets. A shared per-class permutation yields nested support sets. Each local condition contains 20 support seeds and three training seeds, with matched graph, checkpoint, and support/query identities. The pretrained source is the original edge-prediction checkpoint for each dataset; its pretraining seed is unavailable.

The local references are **Frozen**, with no prompt and the same final prototype readout as PRPrompt, and **Full FT**, updating the pretrained encoder and a task readout. Prompt baseline accuracy is taken from Tables 4–6 of Chen et al. [2]: GPPT, Gprompt (GraphPrompt), All-in-one, GPF, and GPF-plus, together with their supervised and fine-tuning references. Their MTG method is excluded. The source combines ProG results with author reproductions and reports best performance across pretraining strategies; the selected strategy is not identified for each cell.

Literature protocol. The source uses a 90% test allocation, five independent task samplings, and random hyperparameter search. Its main backbone is a two-layer GCN with hidden dimension 128. These results are not matched to our test split, checkpoint, or fitting budget. Table 1 therefore separates literature and local panels. We retain the source means and standard deviations without reaggregation, and compute no paired intervals or global ranking across panels. The source tables report Accuracy only; the other metric tables contain local measurements.

PRPrompt and the shared-prompt ablation use one fixed recipe. Local Full FT uses three support-only folds at 3 and 5 shots and four candidates: learning rate in $\{0.001, 0.01\}$ and weight decay in $\{0, 0.0001\}$. Each candidate fit and final fit uses 100 steps; 1-shot uses fixed parameters. Consequently, local methods do not

Table 1: Accuracy on few-shot node classification. Higher is better (↑).

Method	Cora			CiteSeer			PubMed		
	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot
Literature-reported baselines [2]									
Supervised	26.56± 5.55	37.79± 9.16	50.25± 8.37	21.78± 7.32	35.18± 6.86	41.22± 6.30	39.37± 16.34	57.33± 4.64	67.88± 2.18
Fine-tuning	52.61± 1.73	51.97± 2.84	62.66± 3.55	35.05± 4.37	45.08± 2.09	39.54± 3.54	46.74± 14.89	65.40± 3.00	**70.91**± 4.87
GPPT	43.15± 9.44	43.84± 6.11	51.98± 3.43	37.26± 6.17	42.34± 8.31	45.77± 7.41	48.31± 17.72	67.43± 2.96	66.97± 3.70
Gprompt	**56.66**± 11.22	**63.78**± 5.77	**69.03**± 3.61	53.21± 10.94	60.00± 6.18	66.13± 1.64	39.74± 15.35	66.68± 3.53	67.87± 2.08
All-in-one	52.39± 10.17	48.09± 4.83	30.36± 13.48	40.41± 2.80	48.09± 8.18	27.93± 10.59	45.17± 6.45	65.79± 5.79	46.16± 15.83
GPF	38.57± 5.41	34.84± 19.83	35.43± 1.02	31.16± 8.05	25.92± 12.30	25.12± 3.01	**49.99**± 8.86	**71.20**± 2.82	68.96± 3.99
GPF-plus	55.77± 10.30	56.38± 5.37	66.22± 6.20	**59.67**± 11.87	**72.48**± 5.63	**75.73**± 2.19	46.64± 18.97	70.85± 4.03	69.59± 4.33
Local measurements (different protocol)									
Frozen	23.17± 5.48	29.42± 3.47	33.56± 5.03	21.14± 2.87	25.29± 2.16	27.38± 2.84	55.44± 11.42	65.55± 8.11	68.37± 3.41
Full FT	45.41± 10.18	61.54± 6.78	67.50± 5.06	35.76± 6.80	42.22± 5.88	47.89± 4.85	48.77± 8.54	59.75± 7.97	63.96± 5.18
PRPrompt	**54.59**± 9.18	**70.76**± 2.76	**73.02**± 1.95	**47.12**± 6.31	**61.20**± 4.98	**66.17**± 2.16	**58.70**± 9.78	**68.66**± 4.58	**70.98**± 2.46

Mean ± std. (%). Literature panel: Chen et al. [2], Tables 4–6, five trials, 90% test allocation, best reported across pretraining strategies. Local panel: fixed edge-pretrained checkpoint and official test split; three training seeds averaged per support draw, then mean/sample std. over 20 draws. **Bold**/underline: best/second within each panel and column only. Protocols differ; no cross-panel ranking, pairing, or pooled average.

Table 2: Macro-F1 on few-shot node classification. Higher is better (↑).

Method	Cora			CiteSeer			PubMed		
	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot
Frozen	22.51± 4.82	29.07± 3.47	33.07± 4.47	20.39± 2.62	24.32± 1.87	26.45± 2.71	53.44± 10.49	63.19± 8.60	66.27± 3.55
Full FT	44.38± 10.07	60.94± 6.41	66.99± 4.52	30.99± 6.30	38.43± 5.80	44.62± 4.98	44.52± 9.85	57.64± 9.26	62.53± 6.05
PRPrompt	**53.34**± 8.94	**69.38**± 2.91	**72.04**± 1.77	**44.26**± 5.75	**58.19**± 4.67	**63.03**± 2.12	**57.37**± 10.07	**68.21**± 4.08	**70.39**± 2.47

Local runs only: mean ± sample std. over 20 support draws after averaging three training seeds per draw. **Bold**/underline: best/second in each column and metric panel. No cross-condition average. Literature baselines do not report this metric in Tables 4–6.

have identical total optimization budgets. Appendix A gives the remaining settings.

Statistical unit. For a fixed local condition, we first average three training seeds within each support draw, then report the mean and sample standard deviation over the 20 support draws. We never average performance across datasets or shot counts. Local paired intervals resample support draws with 10,000 bootstrap samples. They are unadjusted 95% intervals for the fixed graph and checkpoint, and apply only to Frozen, Full FT, and shared-prompt comparisons. Accuracy and Macro-F1 are the primary reported metrics; the remaining local metrics appear in Appendix B.

5.2 RQ1: Adaptation performance

The local comparison tests adaptation under the same graph, checkpoint, and labeled support budget. Table 1 adds literature context with a separate protocol; Table 2 reports local Macro-F1. Each column is an individual condition, without a pooled average.

On Cora at 5 shots, PRPrompt reaches 73.02% accuracy compared with 33.56% for Frozen and 67.50% for Full FT. On CiteSeer at the same shot count, the corresponding accuracies are 66.17%, 27.38%, and 47.89%. PubMed has a stronger frozen reference: 68.37% at 5 shots, compared with 70.98% for PRPrompt and 63.96% for Full FT. Thus, the improvement over frozen pretraining is substantially smaller on PubMed, an important dependence that a cross-dataset average would hide.

Literature-reported prompt results. The imported values do not support a claim of uniformly higher accuracy. On CiteSeer, GPF-plus reports 59.67, 72.48, and 75.73% at 1, 3, and 5 shots, whereas our corresponding measurements are 47.12, 61.20, and 66.17%. On Cora at 5 shots, our 73.02% lies above the reported Gprompt value of 69.03%; on PubMed at 3 shots, our 68.66% lies below the reported GPF value of 71.20%. These are descriptive cross-source comparisons, not matched estimates of method effects. The evidence for routing comes from the local ablation below; local paired differences are provided in Appendix C.

5.3 RQ2: Value of prototype routing

The central ablation replaces B in Eq. (6) by the constant matrix with entries $1/C$. This retains the prompt

Table 3: Prototype-routing ablation. Matched prompt recipe and readout.

Method	Cora			CiteSeer			PubMed		
	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot
Accuracy (%) ↑									
Frozen	23.17± 5.48	29.42± 3.47	33.56± 5.03	21.14± 2.87	25.29± 2.16	27.38± 2.84	55.44± 11.42	65.55± 8.11	68.37± 3.41
Shared prompt	22.62± 4.41	31.70± 4.95	36.08± 4.19	20.72± 2.46	24.01± 2.70	25.73± 2.78	53.84± 10.55	62.96± 7.94	67.95± 3.11
PRPrompt	**54.59**± 9.18	**70.76**± 2.76	**73.02**± 1.95	**47.12**± 6.31	**61.20**± 4.98	**66.17**± 2.16	**58.70**± 9.78	**68.66**± 4.58	**70.98**± 2.46
Macro-F1 (%) ↑									
Frozen	22.51± 4.82	29.07± 3.47	33.07± 4.47	20.39± 2.62	24.32± 1.87	26.45± 2.71	53.44± 10.49	63.19± 8.60	66.27± 3.55
Shared prompt	21.65± 3.66	30.25± 4.85	34.89± 4.10	19.57± 2.33	22.67± 2.46	24.55± 2.74	51.93± 9.81	60.94± 8.60	66.29± 2.87
PRPrompt	**53.34**± 8.94	**69.38**± 2.91	**72.04**± 1.77	**44.26**± 5.75	**58.19**± 4.67	**63.03**± 2.12	**57.37**± 10.07	**68.21**± 4.08	**70.39**± 2.47

Local runs only: mean ± sample std. over 20 support draws after averaging three training seeds per draw. **Bold**/underline: best/second in each column and metric panel. No cross-condition average. Frozen: no prompt; shared prompt: uniform allocation; PRPrompt: prototype routing.

matrix, scale, injection point, optimization, and final readout, but removes node-dependent allocation. The constant matrix is used directly; centering it would erase the prompt. We also retain the no-prompt Frozen reference.

Table 3 shows that a shared prompt alone does not account for the gains. At 5 shots, routing improves accuracy over the shared prompt by 36.94 percentage points on Cora, 40.45 on CiteSeer, and 3.02 on PubMed. The respective paired 95% intervals are $[34.87, 39.06]$, $[39.06, 41.84]$, and $[1.39, 4.51]$ percentage points. The improvement is also positive in both primary metrics in each 1- and 3-shot condition.

This comparison supports the complete prototype-routing construction relative to a shared prompt. It does not isolate prototype refinement, centering, and graph aggregation as separate causes. A uniform mixture also reduces the effective degrees of freedom despite retaining the nominal prompt parameter count. We therefore do not interpret the comparison as a capacity-matched proof of a unique routing mechanism.

6 Limitations

The evaluation covers three citation graphs, one fixed pretrained checkpoint per graph, and same-graph known-class classification. Pretraining seeds are unknown, so variation across pretraining runs is not estimated. The reporting scope was chosen after the experiments were completed; the analysis is descriptive rather than a new independent confirmatory study. It cannot support claims about graph classification, unseen classes, or arbitrary pretraining sources.

Literature baselines use a different split, five trials, and selection across pretraining strategies. Their aggregate values provide context but cannot isolate an adaptation-method effect against our fixed-checkpoint runs or support paired significance tests. Local fine-tuning selection also incurs a different budget from the fixed PRPrompt recipe. The routing ablation changes effective prompt capacity and does not distinguish every component of the routing construction. Finally, the method uses query features and graph-wide statistics; deployment without such transductive access requires separate evaluation.

7 Conclusion

PRPrompt uses support-anchored prototypes to organize the allocation of trainable hidden-layer prompts in a frozen graph encoder. In the evaluated Cora, CiteSeer, and PubMed settings, condition-wise comparisons and shared-prompt ablations support the value of this allocation. The evidence favors prototype routing as a practical adaptation design within the tested protocol, while leaving its generality across graph families and pretraining sources open.

Broader Impact

The study concerns public citation-network benchmarks and does not evaluate decisions about individuals. Errors can propagate through graph aggregation and unlabeled prototype refinement, particularly when the graph or class structure differs from the evaluated setting. Applications involving consequential decisions require domain-specific evaluation beyond these benchmarks.

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A Reproducibility

A.1 Graph preprocessing and fixed checkpoints

Table 4 records the dataset dimensions from the execution plan. Features use the dataset loader's row normalization. Support pools and query sets come from the official training and test masks; validation labels are not used for fitting or selection. All other nodes remain visible as unlabeled graph context. Each task contains all classes of its graph, with the requested number of support nodes per class.

Table 4: Dataset and prompt sizes.

Dataset	Nodes	Features	Classes	Prompt params.
Cora	2,708	1,433	7	896
CiteSeer	3,327	3,703	6	768
PubMed	19,717	500	3	384

The encoder consists of two GCNConv layers of width 128 with LeakyReLU between them. The first layer includes the pretrained affine transformation and graph aggregation; the prompt is added to its output before the nonlinearity. The second convolution has no additional activation in the evaluated encoder. The same checkpoint is used by every comparison method on a dataset. Checkpoint, graph, and split hashes are preserved in the accompanying result records. The recorded pretraining family is `Edgepred_Gprompt`; the pretraining random seed is unavailable.

A.2 Training and prediction details

The main text describes the active forward computation. For reproducibility, Adam uses $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, learning rate 0.01, and weight decay zero. The prompt regularizer is the mean of squared token entries multiplied by 10^{-3}. It is separate from optimizer weight decay. The channel scale uses the sample standard deviation, with denominator $n-1$. Euclidean normalization uses the implementation's 10^{-12} norm floor. No distillation or query-label objective is active.

For the support loss, let $\bar{z}_i = \mathrm{norm}(Z_i)$ and $m_c = \sum_{j \in S_c} \bar{z}_j$. At $k > 1$, the normalized prototype used for training support node i is

$$\begin{aligned} p_c^{(-i)} &= \mathrm{norm}\left(\frac{m_c - \mathbb{I}[y_i = c]\bar{z}_i}{k - \mathbb{I}[y_i = c]} \right), \\ \ell_{ic} &= \bar{z}_i^\top p_c^{(-i)} / 0.2. \end{aligned} \tag{8}$$

For $k = 1$, the inclusive class prototype $\mathrm{norm}(m_c/k)$ replaces $p_c^{(-i)}$. During prediction, define

$$\begin{aligned} z_i^* &= \mathrm{norm}\left(Z_i - \frac{1}{n} \sum_{j \in V} Z_j \right), \\ p_c^* &= \mathrm{norm}\left(\frac{1}{k} \sum_{i \in S_c} z_i^* \right), \\ \widehat{y}_i &= \arg\max_c (z_i^*)^\top p_c^*. \end{aligned} \tag{9}$$

The probability scores for all five metrics are obtained by applying softmax to cosine logits divided by 0.2. Training does not apply the output centering used at prediction. The readout is a fixed implementation choice, not an experimental factor in this paper.

Procedure. Given a graph, a fixed checkpoint, and support labels:

1. Normalize adjacency and compute Eq. (1) and the first-layer preactivation.
2. Initialize support prototypes with Eq. (2) and apply Eqs. (3)–(4) three times.
3. Compute the fixed route with Eq. (5); for the shared-prompt ablation, replace it by $B_{ic} = 1/C$.
4. Initialize $P = 0$ and cache the hidden-channel scales. Keep the encoder and route fixed.
5. For 100 steps, run the prompted encoder, form support logits, and update P using Eq. (7).
6. Compute final prototype scores using Eq. (9); seal predictions before scoring with query labels.

A.3 Baseline fitting and seed control

The support-draw seeds are 30093000 through 30093019 and training seeds are 0, 1, and 2. A draw uses a seeded permutation in each class's support pool, taking its first k entries. All methods receive the same support identities for each draw. The no-training Frozen method retains the same three seed slots for alignment, but those slots do not represent three independent fits.

Local Full FT uses support cross-entropy and a task readout. Its one-shot parameters are learning rate 0.01 and weight decay 0.0001. For 3 and 5 shots, each of three support-only folds holds out one node per class. Four candidate configurations are ranked by mean held-out support cross-entropy, then Full FT is refit on the complete support set. No external validation-label budget is used. This local protocol controls labeled information and final step count, but not total selection compute or readout parameterization. It does not describe the imported literature baselines.

A.4 Condition-specific statistics

For a fixed dataset, shot count, checkpoint, and method, let a_{st} denote a metric with support seed s and training seed t. We compute

$$\begin{aligned} \bar{a}_s &= \frac{1}{3} \sum_{t=1}^{3} a_{st}, \qquad \widehat{a} = \frac{1}{20} \sum_{s=1}^{20} \bar{a}_s, \\ \widehat{\sigma} &= \sqrt{\frac{1}{19} \sum_{s=1}^{20} (\bar{a}_s - \widehat{a})^2}. \end{aligned} \tag{10}$$

Local cells report $100\widehat{a} \pm 100\widehat{\sigma}$, not the standard error. For local paired comparisons, $d_s = \bar{a}_s^{\mathrm{PRPrompt}} - \bar{a}_s^{\mathrm{reference}}$ is resampled as a support-draw block. The percentile bounds of 10,000 resampled means form the reported 95% interval. Existing intervals are reused without modification. Literature cells retain their source five-trial mean and standard deviation; Eq. (10) does not apply to them. No metric average or mean effect is computed across datasets, shots, methods, or sources.

Accuracy, Macro-F1, Micro-F1, macro one-vs-rest ROC-AUC, and macro average precision are retained in the data package. In this single-label multiclass setting, Micro-F1 coincides with Accuracy; it is included for completeness. The fixed query set is shared across support draws, so intervals concern support selection and training variation conditional on that graph, not independent samples of graphs.

A.5 Environment, provenance, and numerical qualification

The execution plan records Python 3.11.16, PyTorch 2.11.0+cu130, and CUDA 13.0 in the reused ProG environment. Per-unit records identify CUDA devices and fitting durations. An isolated total GPU-hour budget for the present reporting scope and a verified accelerator inventory are not provided by the extracted package; no efficiency comparison is inferred from these results.

Local table entries derive from the sealed three-dataset package. The larger archive contains 3,240 main-comparison units and 540 shared-prompt units; PRPrompt contributes 540 units. This manuscript displays the local Frozen, Full FT, shared-prompt, and PRPrompt subsets. Source numerical audits cover the archived 3,780 units, including paired identities, exact saved-representation readout reproduction, passing forward tolerances, and no changed query classes in fresh forwards. These are archived checks, not new experiments; passing tolerances does not imply bitwise-identical CUDA arithmetic.

External Accuracy values are transcribed from Chen et al. [2], Tables 4–6 (PDF pages 37–38 of arXiv:2606.03290v1). The record preserves table/page identifiers, decimal strings, source URL and PDF hash. It contains 147 node-classification cells excluding MTG; the manuscript uses 63 cells for its three datasets. Source tables combine ProG results and author reproductions, without cell-specific attribution or selected pretraining identifiers. Raw predictions are unavailable in the imported record, so no external paired tests are computed. Local historical prompt measurements remain archived but are not substituted for missing literature metrics.

Source code and the table data are available in the accompanying repository at https://github.com/Mes-sala/PRPrompt. The source revision of the recorded experiments is `0328a2bb68f249a34c89557178b6233daf5ce20d`. The paper package includes table-generation and static-validation scripts, hashes, and bibliographic provenance. Model checkpoints and optimizer states remain on the experiment server and are excluded from the local paper package. This manuscript reports existing experiments; no new training or bootstrap analysis was performed for its preparation.

B Additional Metrics

The following tables extend the main comparison with the remaining stored metrics under the same seed aggregation rule. Every column is a separate dataset/shot condition.

Table 5: Micro-F1 on few-shot node classification. Higher is better (↑).

Method	Cora			CiteSeer			PubMed		
	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot
Frozen	23.17± 5.48	29.42± 3.47	33.56± 5.03	21.14± 2.87	25.29± 2.16	27.38± 2.84	55.44± 11.42	65.55± 8.11	68.37± 3.41
Full FT	45.41± 10.18	61.54± 6.78	67.50± 5.06	35.76± 6.80	42.22± 5.88	47.89± 4.85	48.77± 8.54	59.75± 7.97	63.96± 5.18
PRPrompt	**54.59**± 9.18	**70.76**± 2.76	**73.02**± 1.95	**47.12**± 6.31	**61.20**± 4.98	**66.17**± 2.16	**58.70**± 9.78	**68.66**± 4.58	**70.98**± 2.46

Local runs only: mean ± sample std. over 20 support draws after averaging three training seeds per draw. **Bold**/underline: best/second in each column and metric panel. No cross-condition average. Literature baselines do not report this metric in Tables 4–6.

Table 6: Macro ROC-AUC on few-shot node classification. Higher is better (↑).

Method	Cora			CiteSeer			PubMed		
	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot
Frozen	62.95± 4.93	68.95± 2.99	72.36± 2.85	56.11± 3.24	60.17± 2.06	62.69± 2.35	73.87± 11.03	82.17± 6.59	84.83± 2.59
Full FT	81.83± 5.73	89.94± 2.35	**92.58**± 1.26	70.97± 4.63	76.04± 3.00	79.54± 2.22	70.73± 9.76	81.00± 3.74	84.31± 2.45
PRPrompt	**83.06**± 5.42	**90.84**± 1.97	92.47± 1.11	**74.71**± 5.81	**82.82**± 3.08	**85.86**± 1.52	**76.48**± 10.43	**86.29**± 3.10	**87.03**± 1.87

Local runs only: mean ± sample std. over 20 support draws after averaging three training seeds per draw. **Bold**/underline: best/second in each column and metric panel. No cross-condition average. Literature baselines do not report this metric in Tables 4–6.

Table 7: Macro AP on few-shot node classification. Higher is better (↑).

Method	Cora			CiteSeer			PubMed		
	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot
Frozen	23.64± 4.26	29.20± 3.71	33.16± 4.26	21.53± 2.32	24.17± 1.59	26.46± 2.07	57.52± 10.93	67.27± 8.59	70.34± 4.13
Full FT	54.01± 10.10	70.20± 4.98	75.99± 3.16	38.74± 5.84	45.56± 4.79	51.03± 4.40	55.36± 10.55	67.52± 6.82	72.33± 5.00
PRPrompt	**59.04**± 9.07	**74.33**± 3.38	**77.64**± 1.87	**45.99**± 6.79	**59.15**± 4.80	**64.02**± 2.66	**61.58**± 11.39	**74.22**± 5.19	**75.35**± 3.34

Local runs only: mean ± sample std. over 20 support draws after averaging three training seeds per draw. **Bold**/underline: best/second in each column and metric panel. No cross-condition average. Literature baselines do not report this metric in Tables 4–6.

C Paired Differences and Uncertainty

Each cell reports PRPrompt minus a local reference in percentage points, with its unadjusted 95% paired support-bootstrap interval. All entries preserve original paired statistics. Literature baselines are excluded because the runs are not paired. Tables group conditions for readability but do not average them.

Table 8: Paired Accuracy gains of PRPrompt. Differences in percentage points.

Reference	Cora			CiteSeer			PubMed		
	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot
Frozen	+31.42 [28.20, 34.59]	+41.34 [39.63, 43.04]	+39.46 [37.23, 41.61]	+25.98 [23.72, 28.21]	+35.91 [33.05, 38.39]	+38.78 [37.37, 40.18]	+3.26 [0.40, 6.22]	+3.11 [0.38, 6.27]	+2.61 [1.13, 4.22]
Full FT	+9.17 [6.58, 11.87]	+9.22 [6.75, 11.79]	+5.52 [3.73, 7.50]	+11.36 [9.57, 13.18]	+18.98 [16.78, 20.94]	+18.28 [16.22, 20.13]	+9.93 [6.77, 13.14]	+8.90 [5.89, 12.26]	+7.02 [4.96, 9.14]
Shared prompt	+31.96 [28.59, 35.21]	+39.06 [36.49, 41.44]	+36.94 [34.87, 39.06]	+26.40 [24.07, 28.73]	+37.18 [34.90, 39.33]	+40.45 [39.06, 41.84]	+4.85 [1.26, 8.26]	+5.69 [3.25, 8.27]	+3.02 [1.39, 4.51]

Each cell: PRPrompt minus the reference (top); unadjusted 95% paired support-bootstrap interval (below). All values preserve the original condition-wise statistics.

Table 9: Paired Macro-F1 gains of PRPrompt. Differences in percentage points.

Reference	Cora			CiteSeer			PubMed		
	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot	1-shot	3-shot	5-shot
Frozen	+30.84 [27.77, 33.84]	+40.32 [38.64, 42.08]	+38.97 [36.88, 40.99]	+23.87 [21.80, 25.77]	+33.87 [31.34, 36.14]	+36.58 [35.22, 37.95]	+3.93 [0.86, 7.00]	+5.02 [2.27, 8.44]	+4.12 [2.53, 5.91]
Full FT	+8.97 [6.40, 11.77]	+8.44 [6.12, 10.90]	+5.05 [3.34, 6.86]	+13.27 [11.17, 15.31]	+19.76 [17.45, 22.00]	+18.42 [16.07, 20.60]	+12.86 [9.39, 16.37]	+10.57 [7.08, 14.41]	+7.86 [5.54, 10.28]
Shared prompt	+31.70 [28.29, 35.00]	+39.13 [36.60, 41.48]	+37.15 [35.09, 39.23]	+24.69 [22.70, 26.65]	+35.52 [33.31, 37.55]	+38.48 [37.22, 39.72]	+5.44 [1.84, 9.07]	+7.27 [4.64, 10.29]	+4.10 [2.56, 5.58]

Each cell: PRPrompt minus the reference (top); unadjusted 95% paired support-bootstrap interval (below). All values preserve the original condition-wise statistics.